Module 2 : Semantic Analysis

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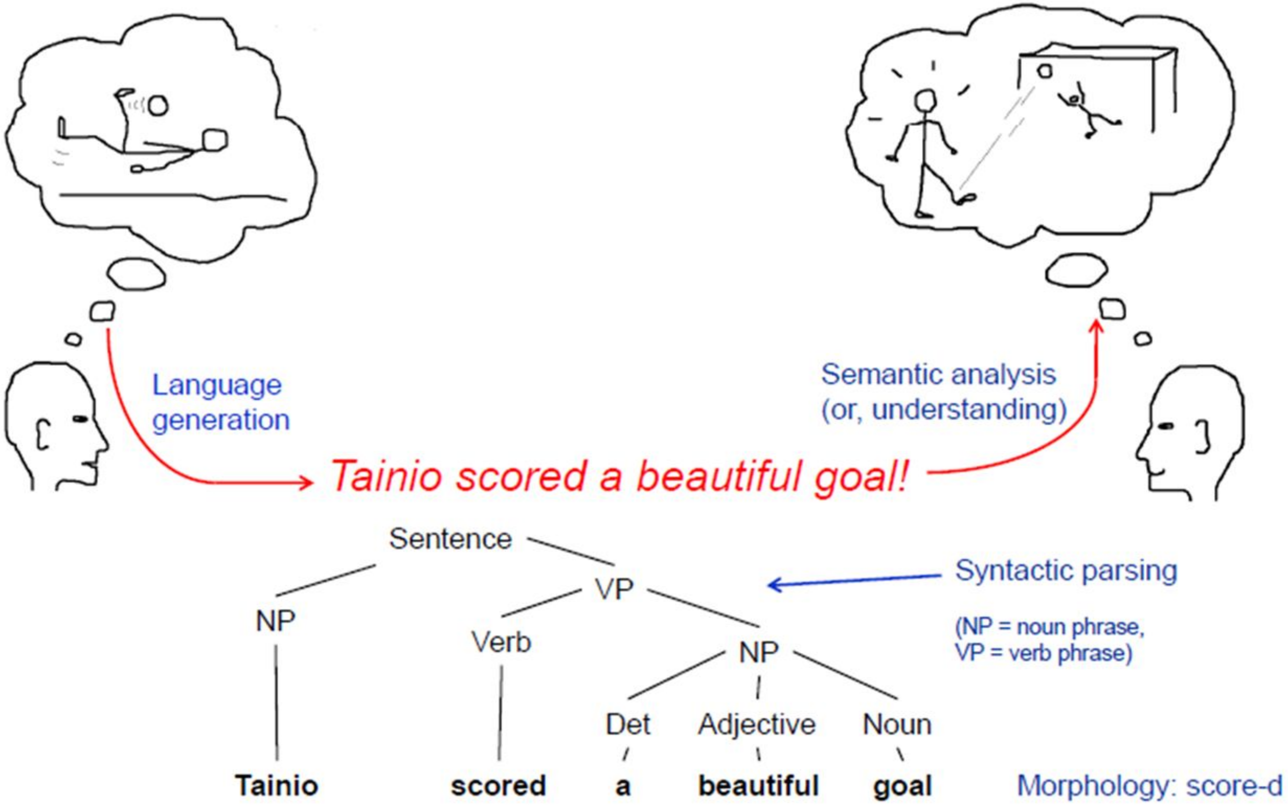
Agenda

- Lexical Semantics
- WordNet Relations among lexemes & their senses
- Homonymy
- Polysemy
- Synonymy
- Hyponymy
- Semantic Ambiguity
- Word Sense Disambiguation (WSD)
- Knowledge based approach (Lesk's Algorithm)

What is Semantic Analysis?



What is Semantic Analysis?

= figure out the meaning of linguistic input (construct meaning representations)

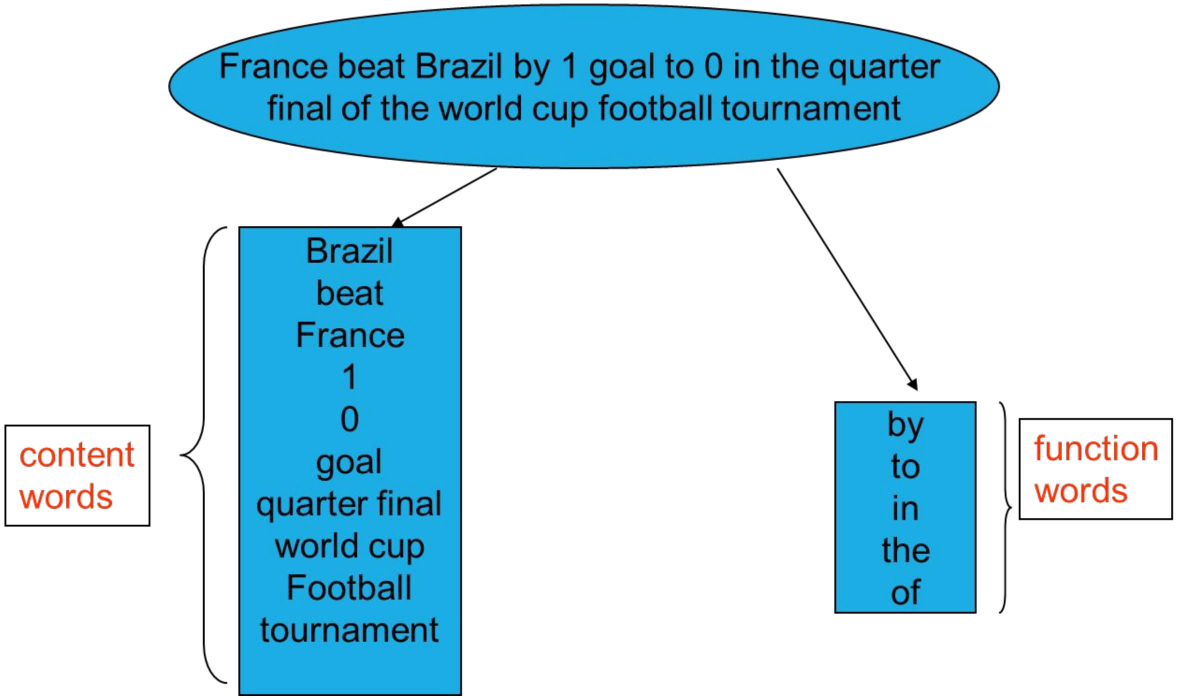
= process language to produce common-sense knowledge about the world (extract data and construct models of the world)

- Lexical semantics
  - meanings of component words
  - word sense disambiguation (e.g. ”country” in political or musical sense)
- Compositional semantics
  - how words combine to form larger meanings

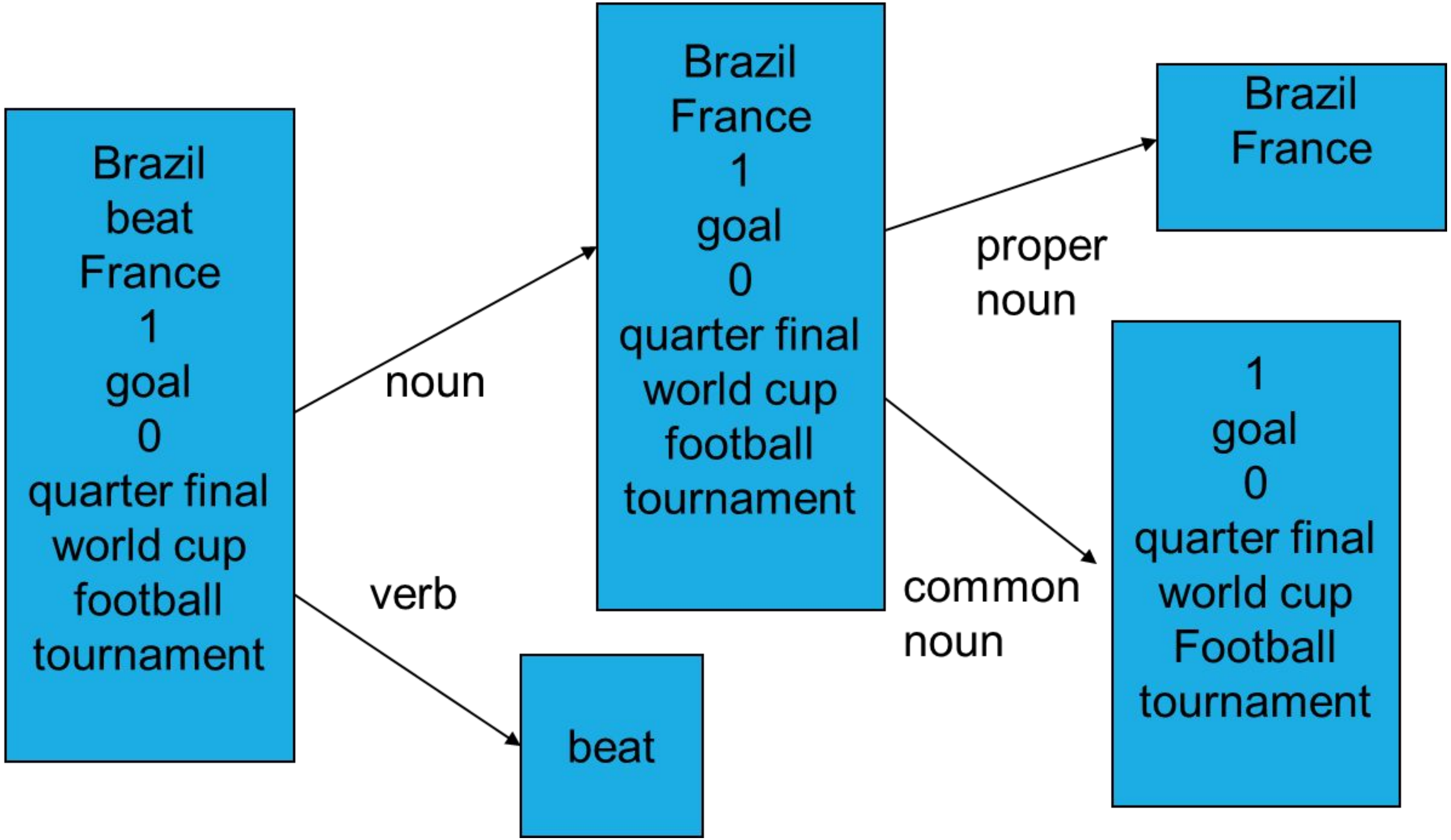
- Roughly: semantic analysis ≈ understanding language

Language Understanding

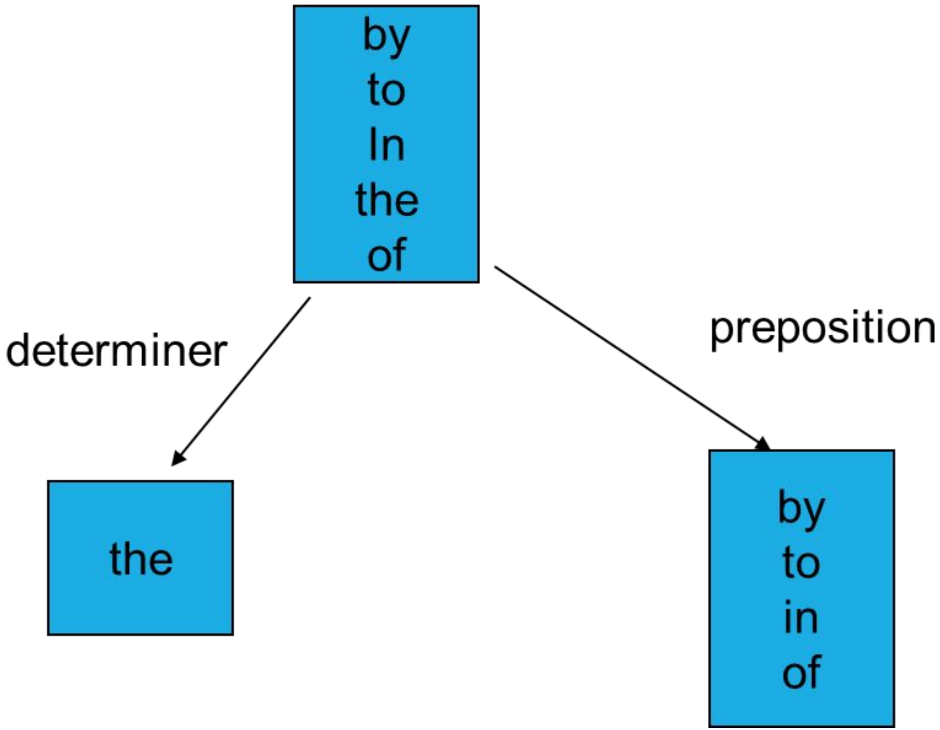
- **Understanding an Utterance** - an uninterrupted chain of spoken or written language. **Semantic :** the study of meaning
- Extracting knowledge about the world from the utterance.



Categories of Words



Categories of Words: Further Classification

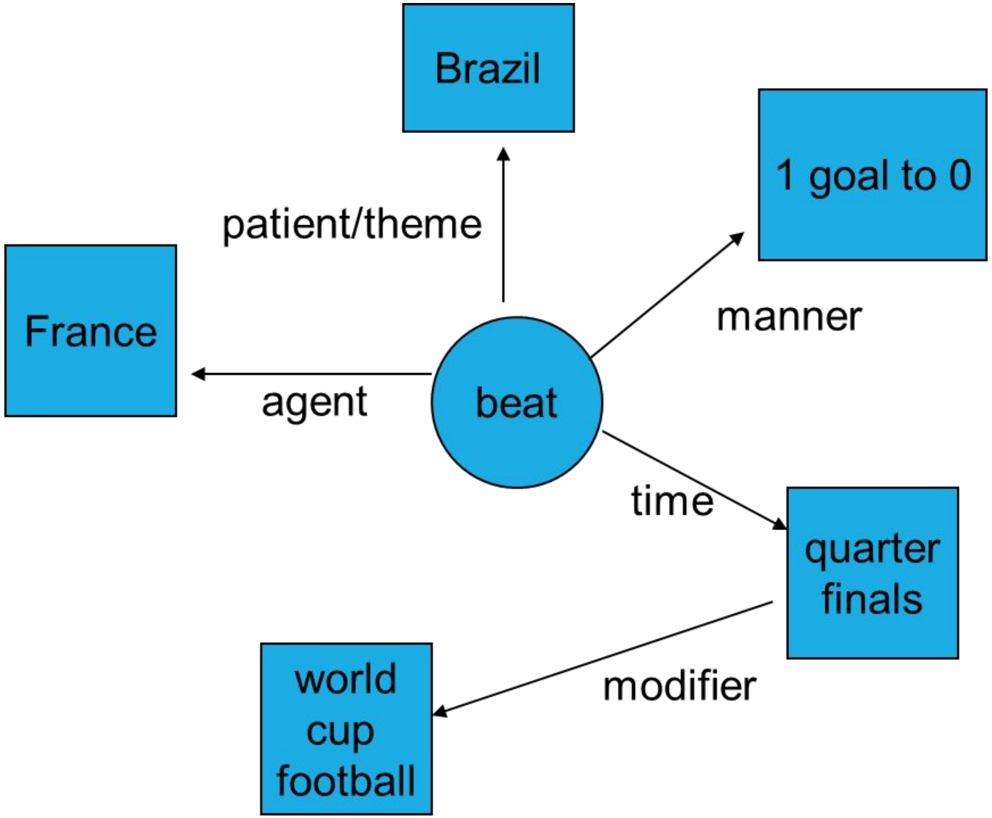


Why all these Categories?

Information need

- who did what
- to whom
- by what
- when
- where
- in what manner

Semantic Roles



Applications of Semantic Analysis

- Information Extraction
- Text Summarization
- Information Retrieval and Document Classification
- Machine Translation
- Human-Computer Interaction : Conversational Agents
- Expert Systems

Lemma and Word Form

- A Lemma or Citation Form
    - Basic part of the word, same stem, rough semantics
- A Word Form
    - The “inflected” word as it appears in text.
- **Eg.:** Word Form : Banks, Lemma : Bank
    - Word Form : Sung, Lemma : Sing

Word Senses

- One Lemma **"bank"** can have many meanings:
  - **Sense 1:** …a bank can hold the investments in a custodial account..
  - **Sense 2:** …as agriculture burgeons on the east bank the river will shrink even more..
- Sense or (Word Sense)
  - A discrete representation of an aspect of a word's meaning
- **Eg.:** In the above example, lemma: bank has 2 senses.

Relations among Lexemes & their Senses

- Homonyms
- Polysemy
- Synonyms
- Antonyms
- Hyponymy and
- Hypernymy

Homonyms

Homonyms: words that share a form but have unrelated, distinct meanings:

bank$_1$: financial institution, bank$_2$: sloping land

bat$_1$: club for hitting a ball, bat$_2$: nocturnal flying mammal

1. Homographs (bank/bank, bat/bat)
2. Homophones:
    1. Write and right
    2. Piece and peace

Homonymy in NLP

- Information Retrieval: “bat care”
- Machine Translation: similar sound
  - bat (animal) or bate (for baseball)
- Text-to-Speech : similar spelling & sound
  - bass (Stringed Instrument) vs. bass (fish)
- Two or more words become homonyms if they either **sound the same (homophones)**, have the **same spelling (homographs)**, or if they **have both homophones and homographs**, but do not have related meanings.

Homonymy in NLP : Examples

- **Homographs** - Sow

  – adult female pig

  – to plant seeds in a ground

- **Homophones**
  - Read vs Reed
  - Right vs Write
  - Pray vs Prey

Polysemy

same meaning but in the same/related context

- Polysemy is a Greek word, which means "**many signs**".
- It is a **word or phrase with different but related sense**.
- In other words, we can say that polysemy has **the same spelling but different and related meaning.**
- For example, the word "**bank**" is a polysemy word having the following meanings –
    - A financial institution.
    - The building in which such an institution is located.
    - A synonym for "to rely on".

Polysemy : Examples

- He drank a glass of milk.
- He forgot to milk the cow.

- The enraged actor sued the newspaper.
- He read the newspaper.

- His cottage is near a small wood.
- The statue was made out of a block of wood.

- He fixed his hair.
- They fixed a date for the wedding.

Difference between Polysemy & Homonymy

- Polysemy and Homonymy are two similar concepts in linguistics.
- Both of them refer to words having multiple meanings.
- **Polysemy refers to the coexistence of many possible meanings for a word or phrase.**
- **Homonymy refers to the existence of two or more words having the same spelling or pronunciation but different meanings and origins.**
- This is the main difference between polysemy and homonymy.

Difference between Polysemy & Homonymy

- Both polysemy and homonymy words have the same syntax or spelling.
- The main difference between them is that **in polysemy, the meanings of the words are related**, but **in homonymy, the meanings of the words are not related.**
- For example, if we talk about the same word "Bank", we can write the meaning 'a financial institution' or 'a river bank'.
- In that case it would be the **example of homonym** because the meanings are unrelated to each other.

Quick Test: Recognize Polysemy & Homonymy

1. a. Sarah climbed down the ladder. b. Sarah bought a down blanket.
2. a. The newspaper got wet in the rain. b. The newspaper fired some of its editing staff.
3. a. My dog would always bark at mailman. b. The tree's bark was a rusty brown
4. a. John was a good man. He donated a lot of money to charity. b. Bill was a good painter. His drawings always were exciting to look at.

Synonyms

- Word that **have the same meaning in some or all contexts**.
  - big / large.
  - automobile / car
  - vomit / throw up
  - couch / sofa
  - water / H2O
- Two words are synonyms, if they can be substituted for each other
- Very few examples of perfect synonymy - often have different notions of politeness, slang etc.
- **Examples are:** ‘author / writer’, ‘fate/destiny’

Synonyms

- Perfect Synonymy is rare
  - Consider this eg.: big / large. - Are they synonyms?
  - *How big is that plane?*
  - *Would I be flying on a large or small plane?*
- Another Eg.:,
  - *Miss Nelson became a kind of big sister to Benjamin*
  - *Miss Nelson became a kind of large sister to Benjamin*
- Why?
  - big has a sense that means being older or grown up
  - large lacks this sense
- **Synonymous relations must be defined between senses.**

Antonyms

- Senses that are **opposites** with respect to one feature of meaning. Otherwise, they are similar!
    - *dark - light, short - long, fast - slow, rise - fall, hot - cold, up - down, in - out*
- Antonyms can
    - define a binary opposition : in /out
    - Be at the opposite ends of a scale : fast / slow
    - Be reversives : rise / fall
- **Very tricky to handle with some representations - remember for later!**

Hyponymy and Hypernymy

- One sense is a **hyponym** of another if the first sense is more specific, denoting a subclass of the other
  - *car* is a hyponym of *vehicle*
  - *mango* is a hyponym of *fruit*
- Conversely **hypernym/superordinate** ("hyper is super")
  - *vehicle* is a hypernym of *car*
  - *fruit* is a hypernym of *mango*
- Usually transitive
  - (A hypo B and B hypo C entails A hypo C)

Superordinate/hyper	vehicle	fruit	furniture
Subordinate/hyponym	car	mango	chair

Word Net

- A hierarchically organized lexical database
- Online thesaurus + aspects of a dictionary
  - *word senses and sense relations*
  - *For other few foreign languages are available*

Category	Unique Strings
Noun	117,798
Verb	11,529
Adjective	22,479
Adverb	4,481

Word Net: Example

https://wordnetcode.princeton.edu/standoff-files/core-wordnet.txt

WordNet Search - 3.1

- WordNet home page - Glossary - Help

Word to search for: bank [Search WordNet]

Display Options: (Select option to change) ▼ [Change]

Key: "S:" = Show Synset (semantic) relations, "W:" = Show Word (lexical) relations

Display options for sense: (frequency) (gloss) "an example sentence"

Noun

- (25)S: (n) **bank** (sloping land (especially the slope beside a body of water)) *"they pulled the canoe up on the bank"; "he sat on the bank of the river and watched the currents"*
- (20)S: (n) depository financial institution, **bank**, banking concern, banking company (a financial institution that accepts deposits and channels the money into lending activities) *"he cashed a check at the bank"; "that bank holds the mortgage on my home"*
- (2)S: (n) **bank** (a long ridge or pile) *"a huge bank of earth"*
- (1)S: (n) **bank** (an arrangement of similar objects in a row or in tiers) *"he operated a bank of switches"*
- S: (n) **bank** (a supply or stock held in reserve for future use (especially in emergencies))
- S: (n) **bank** (the funds held by a gambling house or the dealer in some gambling games) *"he tried to break the bank at Monte Carlo"*
- S: (n) **bank**, cant, camber (a slope in the turn of a road or track; the outside is higher than the inside in order to reduce the effects of centrifugal force)

Open English Wordnet

LEMMA bank

OPTIONS ▼

Nouns

(n) bank *sloping land (especially the slope beside a body of water) "they pulled the canoe up on the bank" "he sat on the bank of the river and watched the currents"*

MORE ▶

(n) bank, banking company, banking concern, depository financial institution *a financial institution that accepts deposits and channels the money into lending activities "he cashed a check at the bank" "that bank holds the mortgage on my home"*

MORE ▶

(n) bank *a long ridge or pile "a huge bank of earth"*

MORE ▶

(n) bank *an arrangement of similar objects in a row or in tiers "he operated a bank of switches"*

MORE ▶

(n) bank *a supply or stock held in reserve for future use (especially in emergencies)*

MORE ▶

(n) bank *the funds held by a gambling house or the dealer in some gambling games "he tried to break the bank at Monte Carlo"*

MORE ▶

https://en-word.net/lemma/bank

Senses and Synsets in WordNet

- Each word in wordnet has at least one sense.
- The Synset (Synonym Set), the set of near synonyms, is a set of senses with a gloss
- **Example:** "Chump" as a noun with the gloss:
  - 'a person who is gullible and easy to take advantage of'
- **This sense of 'Chump' is shared by 7 other words as shown below: All these senses have the same gloss**

Every word has one or more senses.

Each sense corresponds to exactly one synset.

A synset is a set of synonymous words (lemmas) representing the same meaning.

Nouns

oewn-09940867-n (**Interlingual Index:** i88983)

(n) chump, fool, gull, mark, mug, patsy, soft touch, sucker *a person who is gullible and easy to take advantage of*

Senses of “Bass” in WordNet

1. bass - (the lowest part of the musical range)
2. bass, bass part - (the lowest part in polyphonic music)
3. bass, basso - (an adult male singer with the lowest voice)
4. sea bass, bass - (flesh of lean-fleshed saltwater fish of the family Serranidae)
5. freshwater bass, bass - (any of various North American lean-fleshed freshwater fishes especially of the genus Micropterus)
6. bass, bass voice, basso - (the lowest adult male singing voice)
7. bass - (the member with the lowest range of a family of musical instruments)
8. bass - (nontechnical name for any of numerous edible marine and freshwater spiny-finned fishes)

Inventory of Sense Tags for "Bass"

WordNet Sense	Spanish Translation	Roget Category	Target Word in Context
$bass^4$	lubina	FISH/INSECT	…fish as Pacific salmon and striped **bass** and…
$bass^4$	lubina	FISH/INSECT	…produce filets of smoked **bass** or sturgeon…
$bass^7$	bajo	MUSIC	…exciting jazz **bass** player since Ray Brown…
$bass^7$	bajo	MUSIC	…play **bass** because he doesn't have to solo…

WordNet Noun Relations

Relation	Also called	Definition	Example
Hypernym	Superordinate	From concepts to superordinates	*breakfast*^{1} → *meal*^{1}
Hyponym	Subordinate	From concepts to subtypes	*meal*^{1} → *lunch*^{1}
Member Meronym	Has-Member	From groups to their members	*faculty*^{2} → *professor*^{1}
Has-Instance		From concepts to instances of the concept	*composer*^{1} → *Bach*^{1}
Instance		From instances to their concepts	*Austen*^{1} → *author*^{1}
Member Holonym	Member-Of	From members to their groups	*copilot*^{1} → *crew*^{1}
Part Meronym	Has-Part	From wholes to parts	*table*^{2} → *leg*^{3}
Part Holonym	Part-Of	From parts to wholes	*course*^{7} → *meal*^{1}
Antonym		Opposites	*leader*^{1} → *follower*^{1}

Word Sense Disambiguation (WSD)

- Words in natural language usually have a fair number of different possible meanings.
- Ram has a strong interest in computational linguistics.
- Ram pays a large amount of interest on his credit card.
- For many tasks (question answering, translation), the proper sense of each ambiguous word in a sentence must be determined.

Word Sense Disambiguation (WSD)

I play **bass** in a Jazz band

musical_instrument

She was grilling a **bass** on the stove top

freshwater_fish

WSD Methods

- Supervised Machine Learning
- Semi-Supervised Machine Learning
- Thesaurus / Dictionary Methods

WSD Approaches

Knowledge Based Approaches

- Rely on knowledge resources like WordNet, FrameNet etc.
- May use grammar rules for disambiguation.
- May use hand coded rules for disambiguation.

Machine Learning Based Approaches

- Rely on corpus evidence.
- Train a model using tagged or untagged corpus.
- Probabilistic/Statistical models.

Hybrid Approaches

- Use corpus evidence as well as semantic relations form WordNet.

Sample Corpus used in ML Approach

Lexical Sample Task

- Line-hard-serve corpus : 4000 examples of each
- Interest Corpus : 2369 sense-tagged examples.

All Words

- **Semantic Concordance :** a corpus in which each open-class is labeled with a sense from a specific dictionary / thesaurus
- **Semcor :** 234,000 words from Brown Corpus, manually tagged with WordNet Senses.
- **SENSEVAL Competition Corpora :** 2081 tagged word tokens

SemCor : Example

<wf pos=PRP>**He**</wf>

<wf pos=VB lemma=recognize wnsn=4 lexsn=2:31:00::>**recognized**</wf>

<wf pos=DT>**the**</wf>

<wf pos=NN lemma=gesture wnsn=1 lexsn=1:04:00::>**gesture**</wf>

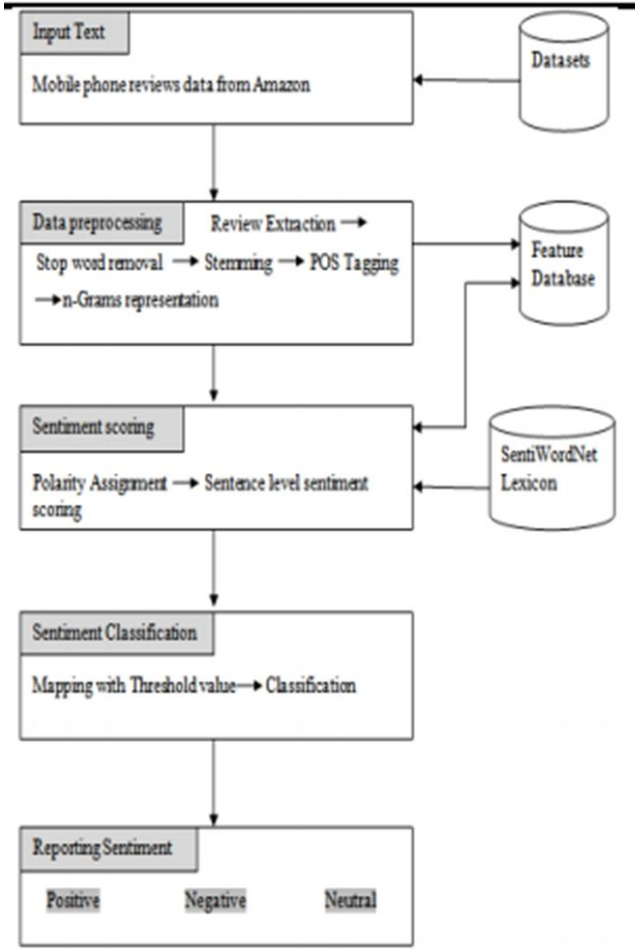
<punc>.</punc>

https://en-word.net/lemma/recognize

Dictionary Based Approach (WSD)

Sentiment Analysis

- **SentiWordNet dictionary** - assigns polarity to each word and then whole sentence
- Predefined threshold is used in classification



Lesk Algorithm

- The Lesk Algorithm (Lesk 1986) is one of the first algorithms for the semantic disambiguation of all words.
- **Required Resource:** a dictionary with definitions (or glosses) for each possible word sense.
- **Idea:** disambiguate words by computing the word overlap among their sense definitions, i.e., word sense whose definition has maximum overlap with the definitions of each context words is chosen.

Lesk Algorithm

Lesk Algorithm:
Pseudocode for disambiguating two words

- for each sense *i* of word W_1
  - for each sense *j* of word W_2
    - compute `overlap(i,j)`, the number of (non-stop) words in common between the definition of sense *i* and sense *j*
- find *i* and *j* for which `overlap(i,j)` is maximized
- assign sense *i* to W_1 and *j* to W_2

Lesk Algorithm : Example

- **Input Sentence:** The **workers** went on **strike** at the **plant.**
- **Step 1: Possible Senses**
  - **Plant (living organism)** - Definition: A living organism such as a tree, shrub, or herb..
  - **Plant (factory)** - Definition: A building or place where industrial work or manufacturing takes place.
- **Step 2: Context words (workers, strike)**
  - **Worker:** A person who works, especially at a job requiring physical labor.
  - **Strike:** A refusal to work organized by employees as a protest.

Lesk Algorithm : Example

- **Step 3: Overlap computation**
- **Sense 1: Living organism**
  - **Words:** living, organism, tree, shrub, herb; **Overlap: none**
- **Sense 2: Factory**
  - **Words:** building, industrial, work, manufacturing
  - **Overlap with:** worker → work; strike → work
    Overlap: work

Step 4: Choose sense

Maximum overlap → **Plant = factory**

Lesk Algorithm : Example

- **Sense Bag:** contains the words in the definition of a candidate sense of the ambiguous word.
- **Context Bag:** contains the words in the definition of each sense of each context word. **E.g.** "On burning **coal** we get **ash**."

Ash

- Sense 1
  Trees of the olive family with pinnate leaves, thin furrowed bark and gray branches.
- Sense 2
  The ***solid*** residue left when ***combustible*** material is thoroughly ***burn***ed or oxidized.
- Sense 3
  To convert into ash

Coal

- Sense 1
  A piece of glowing carbon or ***burn***t wood.
- Sense 2
  charcoal.
- Sense 3
  A black ***solid combustible*** substance formed by the partial decomposition of vegetable matter without free access to air and under the influence of moisture and often increased pressure and temperature that is widely used as a fuel for ***burn***ing

In this case Sense 2 of ash would be the winner sense.

Lesk Algorithm : Example

- **Context given :** E.g. "Pine Cone"
- Dictionary definitions are as follows :

Pine	Cone
Sense 1 Kinds of evergreen tree with needle-shaped leaves.	**Sense 1** Solid body which narrows to a point
Sense 2 Waste away through sorrow or illness.	**Sense 2** something of this shape whether solid or hollow
	Sense 3 Fruit of certain evergreen trees

In this case ???? would be the winner sense.

Lesk Algorithm : Example

- Example context: ***...pine cone...***

	2 word senses of *pine*
#1	kinds of evergreen tree with needle-shaped leaves
#2	waste away through sorrow or illness

	3 word senses of *cone*
#1	solid body which narrows to a point
#2	something of this shape whether solid or hollow
#3	fruit of a certain evergreen tree

Sense inventory contains
2 word senses for *pine* and
3 word senses for *cone*

Lesk Algorithm : Example

- Example context: ***...pine cone...***

2 word senses of *pine*	
#1	kinds of **evergreen tree** with needle-shaped leaves
#2	waste away through sorrow or illness

3 word senses of *cone*	
#1	solid body which narrows to a point
#2	something of this shape whether solid or hollow
#3	fruit of a certain **evergreen tree**

overlap(pine#1, cone#1) = overlap(pine#2, cone#1) = overlap(pine#1, cone#2) = overlap(pine#2, cone#2) = overlap(pine#2, cone#3) = 0

overlap(pine#1, cone#3) = {*evergreen* + *tree*} = 2

→ pine#1 and cone#3 are selected

Lesk Algorithm : Challenges

- When dealing with more than two words, the sense combinations are more complex.
- **Eg.:** "I saw a man who is 108 years old and can still walk and tell jokes".
- There are see (26) × man (11) × year (4) × old (8) × can (5) × still (4) × walk (10) × tell (8) × joke (3), which totals to **43,929,600 sense combinations** (retrieved from WordNet with a part-of-speech tag).
- In order to find the maximum overlapping words, 43,929,600 iterations are needed.
- The time complexity of this algorithm is **exponential.**

Simplified Lesk Algorithm: Solution

- Tries to solve the combinatorial explosion of word sense combinations by disambiguating each ambiguous word in the input text individually, regardless of the meaning of the other words occurring in the same context.
- Correct meaning of each word is determined by finding the sense that leads to the highest overlap between its dictionary definition and the current context.
- Simplified Lesk can significantly outperform the original algorithm in terms of precision and efficiency.

Simplified Lesk only compares the sentence with each sense definition.
It does NOT use definitions of context words (like worker, strike).

Simplified Lesk Algorithm: Pseudocode

- for each sense *i* of *W*
  - determine `overlap(i,c)`, the number of (non-stop) words in common between the definition of sense *i* and current sentential context *c*
- find sense *i* for which `overlap(i,c)` is maximized
- assign sense *i* to *W*

Simplified Lesk Algorithm: Example

- Let's disambiguate "**bank**" in this sentence:

  The **bank** can guarantee deposits will eventually cover future tuition costs because it invests in adjustable-rate mortgage securities.

- given the following two WordNet senses:

$bank^1$	Gloss:	a financial institution that accepts deposits and channels the money into lending activities
	Examples:	"he cashed a check at the bank", "that bank holds the mortgage on my home"
$bank^2$	Gloss:	sloping land (especially the slope beside a body of water)
	Examples:	"they pulled the canoe up on the bank", "he sat on the bank of the river and watched the currents"

Simplified Lesk Algorithm: Example

Choose sense with most word overlap between gloss and context (not counting function words)

The **bank** can guarantee deposits will eventually cover future tuition costs because it invests in adjustable-rate mortgage securities.

bank^1	Gloss:	a financial institution that accepts deposits and channels the money into lending activities
	Examples:	"he cashed a check at the bank", "that bank holds the mortgage on my home"
bank^2	Gloss:	sloping land (especially the slope beside a body of water)
	Examples:	"they pulled the canoe up on the bank", "he sat on the bank of the river and watched the currents"

Simplified Lesk Algorithm: Example

	2 word senses of *bank*
#1	sloping land (especially the slope beside a body of water)
#2	a financial institution that accepts deposits and channels the money into lending activities

Target word: ***bank***

Example context: ***The bank profits from the difference between the level of interest it pays for deposits, and the level of interest it charges in its lending activities.***

Simplified Lesk Algorithm: Example

	2 word senses of *bank*
#1	sloping land (especially the slope beside a body of water)
#2	a financial institution that accepts **deposits** and channels the money into **lending activities**

overlap(bank#1, context) = 0

overlap(bank#2, context)
= {*deposits* + *lending* + *activities*} = 3
→ bank#2 is selected

Example context: ***The bank profits from the difference between the level of interest it pays for deposits, and the level of interest it charges in its lending activities.***